$$
y = \epsilon + \begin{cases} 0, & 0 < t < t_B, \\ \dfrac{Y_{\max}}{1 + \exp\{-k_G(t - t_B - t_0)\}}, & t_B < t < t_B + 2t_0, \\ Y_{\max}, & t_B + 2t_0 < t < t_B + 2t_0 + t_{OP}, \\ SY_{\max} + \dfrac{(1-S)Y_{\max}}{1 + \exp\{k_D(t - t_B - 2t_0 - t_{OP} - t_1)\}}, & t_B + 2t_0 + t_{OP} < t < t_{MAX}. \end{cases}
$$

Symbol	Description
y	Daily step counts at time t
ϵ	Global random noise
t	Time of observation
t_B	Burn-in time during which patient made minimal recoveries following recent op
t_0	Mid-point time of recovery phase after burn-in time ends
k_G, k_D	Logistic growth and decline rate corresponding to model growth and decline phase
$Y_{\max}$	Estimated optimal daily footsteps post-op
t_{OP}	Duration of optimal daily footsteps post-op
S	stagnation in optimal daily footsteps following eg. a possible flare event.
t_1	Mid-point time of flare phase
$t_{\max}$	Maximum time